



University of Hail
College of Computer Science and Engineering
Department of Computer Science
CSCE 242 – Human Computer Interaction
Project Final Report

Haraj APP

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Chapter 1: Introduction

1.1. Project Description

The project focuses on improving the Haraj application by redesigning some usability features to make the buying and selling process easier and more organized for users. The system targets buyers and sellers who use Haraj for product listing, bidding, and communication. The project aims to enhance user experience by improving navigation, bidding, and messaging features.

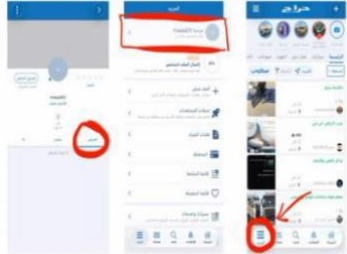
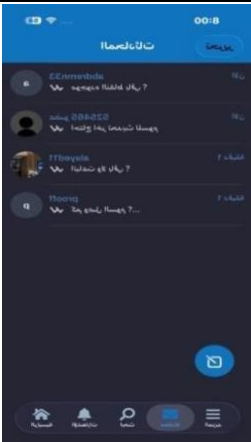
1.2. Goals and Objectives

- Improve the user experience in the Haraj app.
- Simplify the bidding process with a real-time bidding system.
- Make accessing user listings faster and easier.
- Improve communication between buyers and sellers through a product-based messaging interface.
- Reduce navigation complexity and increase usability.

1.3. Identified Problems

- List at least three usability problems you identified in the system, service, technology, or game.
- Explain each problem and the context in which it occurs during use.
- Include screenshots to illustrate these problem

Table 1. Identified Problems

	Problem Title	Description of the Problem	Screenshot of the problem
1	Tracking bids on products	Users find it difficult to track bids clearly during the buying and selling process	
2	Difficulty accessing user listings	Users need multiple steps to access their listings, making navigation less efficient.	
3	Unclear buyer/seller communication	Conversations are not organized by product, causing confusion between buyers and sellers.	

1.4. Target Users

The target users of this project are buyers and sellers who use the Haraj application for online buying, selling, bidding, and communication regarding products. The system is designed for both beginner and experienced users..

Chapter 2: Data Gathering and Analysis

2.1. Data Gathering Methods

2.1.1 Interviews

A total of 17 interviews were conducted. The majority were face-to-face interviews, with two conducted online.

Who was interviewed?

- Current users of the Haraj application (both sellers and buyers)
- Age range: 20–31 years old
- Diverse backgrounds: Cybersecurity, Information Technology, Computer Science, Business Administration, Marketing, Graphic Design, Finance, Human Resources

Questions asked included:

- How do you usually use the Haraj app? (Selling / Buying / Both) ·
- As a seller/buyer, what is the most common problem you face?
- Do you have difficulty tracking the highest bid on a product?
 - Do you find it difficult to access your listed items?
 - Do you face problems organizing or following conversations?

Important feedback, insights, and needs mentioned by users:

- Sellers cannot identify which product a buyer is referring to in messages
 - There is no clear field showing the current highest bid with automatic updates
 - Accessing "My Listings" requires too many steps (More → Profile → Listings)
 - All chats appear in one place without organization or categorization by product
 - New users find it difficult to understand how to list products ·
- Users want verified accounts, escrow payment, and fewer ads

For more details about the interviews, see the Appendix.

2.1.2 Questionnaire/ Survey

A survey was used as the primary quantitative data collection tool. It was designed using Google Forms and included both closed-ended and one open-ended question to evaluate the main usability problems in the Haraj application.

Survey Link:

<https://docs.google.com/forms/d/e/1FAIpQLSdHBtT8Dz89ynQBA2wgrTezxFyXtw74hvuXE5glE5gJN7J47w/viewform>

* Age / العمر

18 - 24 ☒

25 - 34 ☐

35 - 44 ☐

45+ ☐

* Gender / الجنس

Female / أنثى ☒

Male / ذكر ☐

* هل تخصصك أو مجالك مرتبط بالتقنية ؟

Is your major or field related to?

technology

Yes / نعم ☒

No / لا ☐

* كم مرة تستخدم تطبيق حراج ؟

? Haraj app

Figure 1: Screenshot of the survey questions and responses.

I. Pilot study

The names of the group members who evaluated your survey:

From the group "User Experience Evaluation of barq Mobile Application":

1. Hajar Madloul Al-Shammari (202401703)
2. Dareen Adel Al-Furaiddi (202401426)
3. Shorouq Khalaf Al-Mashan (202401485)
4. Saja Badr Al-Rafdi (202401649)

The date of the survey evaluation: April 17, 2026

A brief summary of the feedback received and the improvements made based on the evaluation:

Feedback	Improvement Made
Some questions were compound (contained two ideas)	Split into separate, single-focus questions
Only "Yes/No" options were available	Added "Sometimes" or "Not sure" options
Question "What features would you like to add?" was too general	Added examples (e.g., search, interface, personalization)
Wording of "Are you technical?" was unclear	Rephrased to "Do you have a technical background?"

Figure 2: Screenshot of the evaluation provided by the barq group on our survey.

Evidence of the evaluation your group conducted for another group's survey (Saudi Digital Library):

- **Date:** April 6, 2026
- **Key feedback given:**

- Some questions at the beginning were unclear (e.g., "What is the biggest issue you are currently facing?")
- Question "Are you technical?" is vague → suggested "Do you have a technical background?"
- Section title "Images of the library interfaces" is vague → suggested "Proposed improvements for the Saudi Digital Library interface (with images)"
- Question 18 had too many scattered options → suggested grouping similar options

Figure 3: Screenshots of the feedback provided by our group to the Saudi Digital Library group.

2.2 Data Overview

The survey was distributed through social media (**WhatsApp, Twitter, Telegram**) and university classmates at the University of Hail, as well as friends and relatives.

Total number of responses collected: 71

Target users/participants: Users of the Haraj application, especially individuals with experience in online buying and selling platforms (sellers and buyers).

2.3 Demographic Information

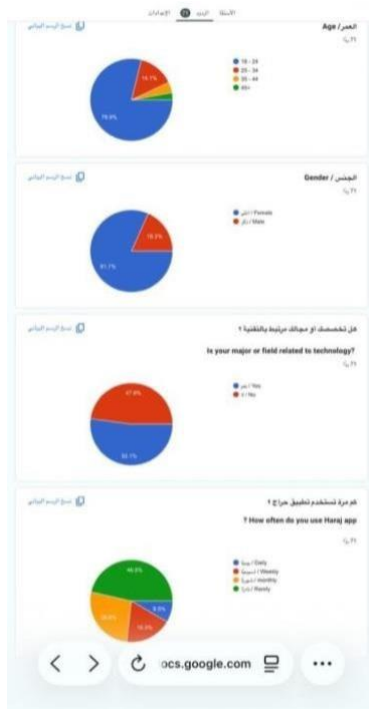


Figure 4: Demographic data of the survey respondents.

Based on the survey data (71 responses):

Age Group	Percentage
18 - 24	78.9%
25 - 34	18.3%
35 - 44	2.8%
+ 45	0%

Gender	Percentage
Female	81.7%
Male	18.3%

Technical Background	Percentage
Has technical background	52.1%
No technical background	47.9%

Summary paragraph of participant characteristics:

The results show that the majority of participants (78.9%) are aged between 18 and 24 years old, which is the most active age group for buying and selling applications. Female participants (81.7%) significantly outnumbered males (18.3%) because the survey was mainly distributed in a female university environment. More than half of the participants (52.1%) have a technical background or interest in technology, meaning they are capable of evaluating usability issues effectively.

Abnormal pattern observed: There is a gender imbalance (mostly female participants). However, this does not affect the validity of the results because the identified problems (bidding, conversations, accessing listings) are common to all users regardless of gender.

2.4 Data Cleaning

No responses were removed for the following reasons:

- All 71 responses were complete (no missing answers on required questions)
- All participants were actual users or had prior experience with Haraj or similar apps
- There were no duplicate or illogical responses

Final number of valid responses after cleaning: 71

2.5 Problems Analysis

Table 1: Problems Analysis

No.	Problem	% Users Agree It Is a Problem	Main Reasons (with %)	Suggested Solutions (with %)	% Users Agree Change Is Needed
1	Difficulty tracking the highest bid (AlSoum) on a product	59.2%	The highest bid is not clearly visible (38.0%) No automatic updates of the latest bid (32.4%)	Add a field under the product showing the highest bid with automatic updates (91.5%) Send push notifications when bid increases (40.0%)	91.5%
2	Difficulty following in-app conversations	59.2%	All chats appear in one place without organization or categorization (42.3%) The messaging interface design does not support easy conversation tracking (32.4%) Difficulty distinguishing between new and old chats (25.4%)	Organize and categorize conversations by product (52.1%) Redesign the messaging interface to make tracking easier (19.7%) Highlight or distinguish new/unread chats (28.2%)	91.5%
3	Difficulty accessing your listed items	52.1%	Too many steps to access listed items (38.0%) No direct shortcut on the main page (35.2%) Unclear interface design (26.6%)	Add a central "My Listings" dashboard to view, edit, delete, and track latest bids (81.7%) Add a direct shortcut to listings on the home page (50.0%)	81.7%

2.6 Discussion

The results presented in Table 1 reveal three major usability problems in the Haraj app, each affecting more than half of the users.

Problem 1 (Difficulty tracking the highest bid) is reported by 59.2% of users. The main reasons are that the highest bid is not clearly visible (38.0%) and there are no automatic updates (32.4%). This suggests that the current bidding interface is static and poorly designed, causing users to miss important bid changes. A large majority (91.5%) believe that adding a field under the product with automatic updates would solve this problem.

Problem 2 (Difficulty following conversations) is also reported by 59.2% of users. The primary reason is that all chats appear in one place without organization (42.3%), followed by poor interface design (32.4%) and difficulty distinguishing new from old chats (25.4%). This indicates that the messaging system lacks basic chat management features. Most users (52.1%) prefer organizing and categorizing conversations by product as the most effective solution. Additionally, 91.5% of users agreed that improving the messaging system would make communication easier overall.

Problem 3 (Difficulty accessing listed items) is reported by 52.1% of users. The main causes are too many steps to access listings (38.0%) and no direct shortcut on the main page (35.2%). This means that sellers waste time navigating through the app. The proposed solution of a central "My Listings" dashboard is preferred by 81.7% of users.

Overall conclusion: The findings show that Haraj app users need better visibility of bids, organized conversations, and faster access to their listings. Implementing these improvements would significantly enhance user experience and satisfaction.

2.7 Analysis of Open-Ended Responses

The survey included one open-ended question:

"Do you have any additional suggestions to improve the user experience of the Haraj app?"

Result: No written responses were provided by participants. All participants left this field blank.

Therefore, no thematic analysis could be performed.

Chapter 3: System Design

3.1 Design Implications (Based on Analysis)

This section explains how the findings from Chapter 2 were used to improve the design of the Haraj application. The identified usability problems were analyzed based on user feedback and questionnaires. Appropriate design decisions were then proposed to solve these issues and improve the overall user experience *Table 2: Design Implications*

No.	Problem	Evidence (from data)	Design Decision	Justification
1	Difficulty tracking bids on products.	Users reported that there is no clear field showing the latest bid amount or bidder information, making the bidding process confusing	Add a bid field for each product showing the latest bid amount and bidder name	This improvement increases transparency in the bidding process and helps users easily track current offers.
2	Accessing user listings requires multiple navigation steps.	Users stated that accessing listings through “More → Profile → Listings” takes too much time and effort.	Create a centralized “My Listings” dashboard with direct access from the home page.	Reducing navigation steps improves usability and allows users to manage their products more efficiently.
3	Communication between buyers and sellers is unclear.	Users mentioned that messages are scattered and do not clearly indicate which product is being discussed	Design a centralized messaging interface linked to each product with notifications.	Organizing conversations by product improves communication clarity and reduces confusion between users.
4	Product status after sale is unclear.	Users reported that sold products do not clearly display whether the item was sold or what the final price was	Add product status labels such as “Sold” and display the final selling price.	Showing the product status improves transparency and helps users understand product availability

3.2 Design Sketches (Low-Fidelity Prototype)

The low-fidelity prototype was created based on the problems identified during the analysis phase. The sketches focus on improving the user experience in the Haraj application by simplifying navigation, improving bidding visibility, organizing communication, and making product status clearer. Ensure the prototype aligns with the previously identified issues and design decisions.

The prototype includes the following main screens:

- Home Screen
- Product Details Screen
- Messaging Screen
- My Listings Dashboard

The sketches were hand-drawn to demonstrate the basic structure and interaction flow before developing the digital prototype.

Low-Fidelity Prototype Screens

Figure 1: Redesigned Home Screen

Description: The redesigned home screen provides quick access to important sections such as “My Listings,” “Messages,” and “Saved Items.” The layout is simplified to reduce clutter and improve navigation.

Figure 2: Product Details Screen with Bid Field

Description: The product details screen includes a new bidding section showing the latest bid amount and bidder information. Users can place or update their bids directly from the product page.

Figure 3: Central Messaging Interface

Description: The messaging interface organizes conversations by product. Each chat clearly shows the related product image and title to reduce confusion between buyers and sellers.

Figure 4: My Listings Dashboard

Description: The “My Listings” dashboard allows users to manage all their products from one place. Users can edit, delete, or update products and view the latest bids easily.

Comparison Between Old and Redesigned System

Old System Problems

- No clear bidding information.
- Difficult access to user listings.
- Messages are disorganized and unclear.
- Sold products do not display their final status.

Redesigned System Improvements

- Added a visible bidding section for products.
- Added direct access to “My Listings.”
- Organized messages by product with notifications.
- Added “Sold” labels and final sale prices.

3.3 Design Sketches (High-Fidelity Prototype)

The high-fidelity prototype was digitally designed to provide a realistic representation of the improved Haraj application. The design was developed based on the usability issues identified from user feedback and analysis results.

The prototype focuses on:

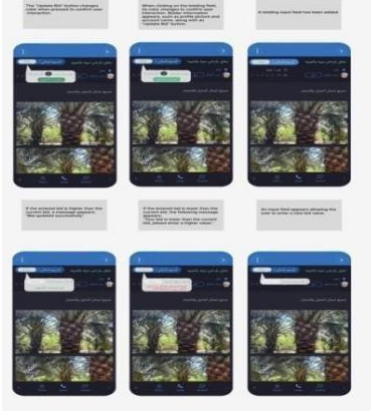
- Improving the bidding system
- Simplifying navigation
- Organizing communication
- Increasing transparency for product status

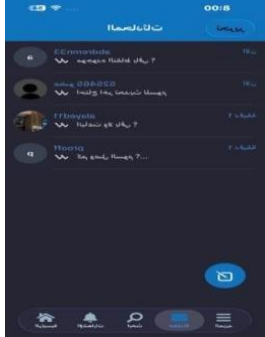
The prototype includes all main user interface elements and redesigned screens The high-fidelity prototype was designed using Figma.

Figma Prototype Link:

(<https://www.figma.com/design/U7RFeN4dxnDmG54v6n8A4a/Untitled?node-id=1-9&t=Ds7civAMFDmRQaJc>)

Table 3 Problem–Solution Mapping with Before and After Design Improvements.

Problem	Before (Issue Screenshot)	Design Solution	After (Improved Screenshot)	Description
Difficulty tracking bids		Added bid field displaying latest bid and bidder information		The new design allows users to easily monitor bidding activity and increases transparency

Accessing listings requires many steps		Added centralized “My Listings” dashboard on home screen		users can now manage their listings quickly without navigating through multiple pages.
Messages are unclear and disorganized		Added product linked messaging interface with notifications		Conversations are now organized by product, making communication clearer and easier

High-Fidelity Prototype Screen Descriptions

Figure 5: High-Fidelity Home Screen

Description: The home screen uses a clean modern layout with quick navigation shortcuts and simplified categories to improve accessibility.

Figure 6: High-Fidelity Product Details Screen

Description: The redesigned product page highlights bidding information, including the latest bid amount and bidder details, making the process more transparent.

Figure 7: High-Fidelity Messaging Interface

Description: The messaging interface displays conversations grouped by product with notifications and product previews for easier communication.

Figure 8: High-Fidelity My Listings Dashboard

Description: The dashboard allows users to manage listings efficiently by editing products, checking bids, and viewing product status from one screen.

Chapter 4: Conclusion and Future Works

4.1. Conclusion

This project aimed to improve the user experience of the Haraj application by addressing three main usability problems: difficulty tracking bids, unclear conversations, and difficulty accessing user listings. Data was collected through interviews and a survey distributed to 71 users. Based on the results, several design improvements were proposed, including a clearer bidding system, organized product-based messaging, and a direct “My Listings” dashboard. These improvements were presented through low-fidelity and high-fidelity prototypes to create a more organized and user-friendly experience for buyers and sellers.

4.2. Future Work

Future improvements may include:

- Adding a secure payment system
- Seller verification badges
- Ad-free premium mode
- Usability testing with real users
- Smart notifications
- Image-based product search
- Improving application performance and loading speed

Chapter 5 : references

- [1] University of Hail, CSCE 242 – Human Computer Interaction Course Materials, College of Computer Science and Engineering, 2026.
- [2] Haraj Application.
- [3] Figma Design Tool.
- [4] Google Forms Survey Tool.

Appendix

Interviews

Interviewee Name: Wasan Al-Gharbi

Interview 1

Interviewee Name: Munirah Salem

Date of Interview: 13 April 2026

Method: Face-to-face

Q1: How do you usually use the Haraj app? (Selling / Buying / Both) A: I only use it for selling.

Q2: As a seller, what is the most common problem you face?

A: When I list multiple products, buyers send me messages asking for more details, but I cannot identify which product they are referring to.

Q3: Would you prefer conversations to be linked to each product? Why?

A: Yes, I would prefer that because it would make things much easier, help me respond faster, and clearly identify which product is being discussed.

Interview 2

Interviewee Name: Faris Algharbi

Date of Interview: 15 April 2026

Method: Face-to-face

Q1: How often do you use the app per week?

A: I rarely use it. I only open it when I post an item for sale to check the current highest bid.

Q2: Do you face any issues tracking the highest bid?

A: Yes, I have to manually update the highest bid, which requires continuous monitoring. Otherwise, the buyer has to search for the latest bid in the messages under the product, but these messages appear randomly. Sometimes buyers ask the seller directly, and I receive many repetitive messages asking about the same thing, which becomes overwhelming.

Q3: Have you ever felt lost while using the app or couldn't find something? Explain.

A: Yes, when I first used the app, I did not know how to list my products. I had to ask a friend who had more experience to guide me.

Q4: Would having a dedicated "My Listings" page in the bottom navigation bar make it easier to use?

A: Yes, it would make things much easier, especially for new users.

Interview 3

Interviewee Name: Khalid Saad

Date of Interview: 15 April 2026

Method: Face-to-face

Q1: How would you rate the overall ease of use of the app? A: It is moderately easy to use.

Q2: What is the most frustrating thing about the app?

A: The interface is not well organized, and it is sometimes difficult to quickly access features.

Q3: Would having a dedicated "My Listings" page in the bottom navigation bar make it easier to use? Why?

A: Yes, because it would allow quick access to all my listings without having to search within the app, and it would save time, especially when I have multiple products.

Interviewee Name: Rana Mohammed

Interview 1

Name: Lamia

Age: 28

Major: Cybersecurity

Interview Type: In-person

Date: 15/4/2026

Q: How do you identify a scam seller?

A: I check their account history and reviews.

Q: What security solution would you like?

A: Escrow payment, where the money goes to the app first.

Interview 2

Name: Abdulaziz

Age: 26

Major: Information Technology

Interview Type: Online

Date: 14/4/2026

Q: How have ads affected you?

A: There are too many, and they make me leave the app by mistake.

Q: What solution do you suggest? A: An ad-free mode with a small subscription fee.

Interview 3

Name: Manal

Age: 30

Major: Computer Science

Interview Type: In-person

Date: 13/4/2026

Q: How was your experience with customer support? A: Bad, they responded after a week and didn't solve the issue.

Q: What would you like instead?

A: A live chat that responds within minutes.

Interview 4

Name: Aljawhara

Age: 24

Major: Information Systems

Interview Type: In-person

Date: 12/4/2026

Q: Does the search give you relevant results?

A: No, it shows unrelated items.

Q: What feature would you like?

A: Image search, where I can take a picture of a product and find similar ones.

Interview 5

Name: Ahmed

Age: 25

Major: Software Engineering

Interview Type: Online

Date: 11/4/2026

Q: Is the app slow for you?

A: Yes, especially when I search or open images.

Q: What technical solution do you suggest?

A: Progressive image loading instead of loading all images at once.

Interviewee Name: Khdaejih Mubarak

Interview 1

Name: Haya

Age: 21

Major: Computer Science

Q: Do you use buying and selling apps like Haraj?

A: Yes, I use them sometimes to buy simple things.

Q: How is your experience?

A: It's okay, but sometimes conversations get lost if I talk to more than one person.

Q: Do you think linking each conversation to a specific product would help? A:

Definitely, it would help me know which conversation is about which product.

Interview 2

Name: Nora

Age: 23

Major: Information Systems

Q: How often do you use the app? A:

About once or twice a week.

Q: Have you faced any issues using it?

A: Yes, especially with organizing messages.

Q: What solution do you suggest?

A: Having a separate conversation for each product.

Interview 3

Name: Mohammed

Age: 22

Major: Software Engineering

Q: Do you sell or buy through the app? A:

Both, but I mostly sell.

Q: Do you find it difficult to reply to customers?

A: Yes, especially when they ask about multiple products at the same time.

Q: Is organizing conversations by product useful?

A: Very useful, it would make replying and tracking much easier.

Interview 4

Name: Reem

Age: 20

Major: Information Technology

Q: How was your first experience with the app?

A: It was a bit confusing.

Q: Why?

A: It wasn't clear how to keep track of conversations.

Q: What would you like to be improved?

A: A simpler interface and better organization.

Interviewee Name: Yara Alharbi

Interview 1

Name: Sara

Age: 27

Major: Business Administration

Q: Do you trust sellers on these apps?

A: Not always, some accounts seem suspicious.

Q: Why do you feel that way?

A: Because some profiles don't have enough information or reviews.

Q: What would increase your trust?

A: Verified accounts with clear badges.

Interview 2

Name: Saad Alharbi

Age: 30

Major: Marketing

Q: What do you think about product descriptions?

A: Sometimes they are unclear or incomplete.

Q: How does that affect you?

A: It makes me hesitate before buying.

Q: What would you suggest?

A: Adding detailed descriptions with clear images.

Interview 3

Name: Abeer

Age: 24

Major: Graphic Design

Q: Do you care about the app design?

A: Yes, design makes a big difference.

Q: How so?

A: A clean design makes it easier to use.

Q: What would you improve?

A: Better colors and clearer layout.

Interview 4

Name: Faisal

Age: 31

Major: Finance

Q: Do you compare prices between sellers? A:

Yes, always before buying.

Q: Is it easy to compare?

A: Not really, it takes time.

Q: What feature would help?

A: A price comparison tool.

Interview 5

Name: Dana

Age: 22

Major: Human Resources

Q: Do notifications help you?

A: Sometimes, but there are too many.

Q: What's the problem with that?

A: I ignore most of them.

Q: What would you prefer?

A: Smart notifications based on my interests.